

Experiment #5

Francisco Gabriel Murillo Roldán

Determination of the Braking Force Limit: A
Case of Experimental Reorientation Following
Unexpected Results

Mining Engineer

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1. Introduction

Experiment #1 (brake shoe tests, April 2026) revealed a paradoxical behaviour: the theoretically more powerful shoe (`High\_Friction\_Composite`) produced worse braking distances than `Disc\_Pads` due to greater intervention of the WSP system. That finding prompted the central question of this experiment: what is the maximum braking force that can be applied without causing wheel lock?

The initial objective was to determine the brake cylinder pressure at which wheel lock occurs for different shoe types. However, the results obtained forced a complete reorientation of the experiment, as documented in this report. Due to technical limitations of the available equipment (inability to dump DataLogger data), the quantitative measurement of braking force is postponed to Experiment #5b, to be carried out with a more powerful computer.

2. Materials

Element	Detail
Locomotives	RENFE 333.4 (120t, CK 0.33/0.03/0.16/1.00) and RENFE 321 (102 t, VK 0.35/0.03/0.132/1.00)
Route	CTN test route, level circuit
Conditions	Dry, MaxWindSpeedMps = 0
Open Rails versions	NYMG 171.1, NYMG 171.3, OR official 1.6.1
Tools	HUD (cylinder pressure, speed), logs

3. Empirical Method

3.1 Original design

The initial plan was to measure the brake cylinder pressure at the moment of wheel lock for three shoe types (Cast_Iron_P6, Cast_Iron_P10, High_Friction_Composite) at speeds of 40, 60, 80, 100 and 120 km/h, with the WSP system deactivated to observe pure wheel lock behaviour.

3.2 Initial results and deviation from the plan

The first tests with locomotive 333.4 produced no wheel lock under any condition. Three different configurations were tested (calibrated with `ORTSMaxBrakeShoeForce`, calibrated with `MaxBrakeForce`, and factory original) without success. Locomotive 321, with the more aggressive Cast_Iron_P6 shoe, also failed to lock initially.

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3.3 Discovery of the key variable

Comparison between configurations revealed that wheel lock depended on the type of parameter used to limit braking force:

Parameter	Value (kN)	Wheel lock?
MaxBrakeForce	500.0	 No
ORTSMaxBrakeShoeForce	750.2	 Yes

ORTSMaxBrakeShoeForce generates a higher braking force at the rail than `MaxBrakeForce` for the same nominal value, because the former is multiplied by the brake shoe friction coefficient.

3.4 Multi-version testing

Three versions of Open Rails were tested with the locomotive 321 configuration (`ORTSMaxBrakeShoeForce` = 750.2 kN, Cast_Iron_P6, CK 0.35/0.03/0.132/1.00, WSP deactivated):

Version	40 km/h	100 km/h
NYMG 171.1	 Wheel lock	 Wheel lock
NYMG 171.3	 Wheel lock	-
OR 1.6.1 official	 No wheel lock	 No wheel lock

3.5 Reorientation of the experiment

Given the impossibility of measuring the locking pressure in current versions of the simulator, the experiment was reoriented towards a more fundamental objective: determining the actual braking force generated by `ORTSMaxBrakeShoeForce` and comparing it with the available adhesion. The quantitative measurement of this force, which requires dumping DataLogger data, is postponed to Experiment #5b due to technical limitations of the current equipment.

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4. Experimental Results

4.1 Wheel lock tests with different configurations

Locomotive	OR Version	Limiting parameter	Shoe	40 km/h	100 km/h
333.4	171.3	`ORTSMaxBrakeShoeForce` 882.6 kN	Composite	✗	✗
333.4	171.3	`MaxBrakeForce` 300 kN	Composite	✗	-
321	171.3	`MaxBrakeForce` 500 kN	P6	✗	✗
321	171.3	`ORTSMaxBrakeShoeForce` 750.2 kN	P6	✓	-
321	171.3	`ORTSMaxBrakeShoeForce` 750.2 kN	P6	✓	✓
321	1.6.1	`ORTSMaxBrakeShoeForce` 750.2 kN	P6	✗	✗

4.2 Log analysis: the zeroadhesion parameter

The Open Rails log reveals that the simulator applies a specific adhesion coefficient for force calculations:

...

Zero Adhesion - zeroadhesion 0.2156863 maxForcekN 220 Driveweight 102000 (321)

Zero Adhesion - zeroadhesion 0.2841667 maxForcekN 341 Driveweight 120000 (333.3)

...

This value varies between locomotives and is systematically lower than parameter A of the CK model. It is not a universal constant, but a value calculated by Open Rails for each locomotive based on its characteristics.

4.3 Additional observation

In the tests where wheel lock occurred, it was observed that the brake cylinder pressure reached its maximum value well before locking took place. This confirms that pressure is not the limiting factor: the resulting braking force is.

5. Analysis

5.1 Why did Experiment #1 show wheel lock?

Experiment #1 (April 2026) was carried out with OR version NYMG 171.1, which did not apply zeroadhesion as a limit to `ORTSMaxBrakeShoeForce`. The combination of a high clamping force (882.6 kN on the 333.4) and a high-friction shoe (Composite) produced braking forces exceeding the available adhesion, causing the observed wheel lock.

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5.2 Why do current versions not lock?

In NYMG versions 171.1 and 171.3, `ORTSMaxBrakeShoeForce` is not limited by zeroadhesion, allowing the braking force to exceed the available adhesion and produce wheel lock. In the official version 1.6.1, zeroadhesion is also applied to `ORTSMaxBrakeShoeForce`, preventing wheel lock.

5.3 Difference between the MG and official branches

Branch	`ORTSMaxBrakeShoeForce` limited by zeroadhesion	Wheel lock in dry?
NYMG 171.1	 No	 Yes
NYMG 171.3	 No	 Yes
OR 1.6.1 official	 Yes	 No

The MG branch (NewYear MG) behaves differently from the official version regarding the braking model. This finding is relevant for any user employing `ORTSMaxBrakeShoeForce` in their calibrations.

5.4 Zeroadhesion is not a fixed value

Comparison between the logs of the 321 and 333.3 demonstrates that zeroadhesion varies between locomotives. It is not a universal constant, but a value calculated by Open Rails for each locomotive based on its characteristics (CK, weight, MaxForce, etc.). This discovery is relevant for understanding brake behaviour in different machines.

6. Conclusions

#	Conclusion	Evidence
1	Wheel lock depends on `ORTSMaxBrakeShoeForce`, not on cylinder pressure	With `MaxBrakeForce = 500 kN` there is no lock; with `ORTSMaxBrakeShoeForce = 750 kN` there is.
2	Zeroadhesion varies between locomotives	0.2157 for the 321, 0.2842 for the 333.3.
3	NYMG versions allow `ORTSMaxBrakeShoeForce` to exceed zeroadhesion	171.1 and 171.3 lock; official 1.6.1 does not.
4	Experiment #1 was made possible by version 171.1	Without the zeroadhesion limit, 882.6 kN of clamping force produced wheel lock.
5	Cylinder pressure is not a relevant variable for wheel lock	It reaches its maximum well before locking occurs.
6	The experiment had to be reoriented in light of the evidence	From measuring locking pressure to investigating the difference between braking parameters and OR versions.
7	Quantitative measurement is postponed to Experiment #5b	Due to technical limitations of the current equipment (no DataLogger data dump capability).

7. Recommendation for the Teluroncio Method

To prevent wheel lock in NYMG versions of Open Rails, `ORTSMaxBrakeShoeForce` should be adjusted so that the resulting braking force does not exceed the product $\text{zeroadhesion} \times \text{weight} \times g$. Since zeroadhesion varies between locomotives, it is recommended to extract this value from each machine's log and calculate the specific limit. It is also recommended to verify this limit on the version of Open Rails to be used, since behaviour differs between the MG and official branches.

8. Future Work

An Experiment #5b is planned which, using a computer with greater processing capacity and the Open Rails DataLogger, will precisely measure the braking force at the exact moment of wheel lock. This will enable quantitative values to be established for the adjustment of `ORTSMaxBrakeShoeForce`.

9. References

- Open Rails Team. *Open Rails 1.6.1 Manual*, Release 1.6.1.0 (2026).
- Yuan, Z., Wu, M., Tian, C., Zhou, J., & Chen, C. (2021). *A Review on the Application of Friction Models in Wheel-Rail Adhesion Calculation*. Urban Rail Transit, 7(1), 1–11.
- Open Rails NewYear MG. Versions 171.1, 171.3. Available at Elvas Tower.
- Technical datasheets RENFE 321 and 333.4. Available at the renfe-calibracion repository.

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